

RESTAURANT ANALYSIS REPORT

Dataset: 9,551 restaurant records

Analysis Focus: Restaurant distribution, ratings, pricing and service availability

Tool Used: Microsoft Excel

1. Project Overview

This project analyzes restaurant data to identify patterns and insights related to **restaurant distribution, ratings, pricing and service availability**.

The project follows a structured Excel-based workflow, beginning with data preparation and quality checks and continuing through exploratory data analysis, PivotTable analysis and dashboard development.

The analysis focuses primarily on the **Indian restaurant market**, while also providing a broader view of the countries represented in the dataset.

Tool Used: Microsoft Excel

Main Objectives:

- Understand restaurant distribution across countries and cities
 - Analyze restaurant ratings and rating categories
 - Examine price-range distribution
 - Compare service availability such as online delivery and table booking
 - Present key findings through an interactive Excel dashboard
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2. Dataset Overview

The project dataset was provided as **multiple separate Excel workbooks**.

The source files consisted of:

- **6 regional restaurant data workbooks**
- **2 supporting workbooks**

The regional files contained restaurant information for different regions, while the supporting files were used during data preparation and validation.

After the data from these sources was consolidated and prepared, the final **Clean Data table** contained:

- **9,551 restaurant records**
- **18 analysis columns**

The final Clean Data table was used as the base dataset for the analysis performed in the separate Analysis workbook.

3. Data Preparation and Quality Checks

Data preparation was performed before starting the analysis to ensure that the final dataset was suitable for use.

The process included:

- Bringing the regional and supporting data into the **Data Preparation workbook**
- Performing **data quality checks** for duplicates and missing values
- Checking **Restaurant IDs and Country Codes**
- Validating **Country Code relationships** using the Country Master
- Checking relevant numeric and categorical fields
- Identifying and addressing data quality issues
- Merging the cleaned regional data into the final **Clean Data table**

After the preparation and quality-checking stage, the cleaned dataset was used in a **separate Analysis workbook** for:

- EDA
 - Insights
 - PivotTables
 - Dashboard
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4. Country Distribution

- India dominates the dataset, with 8,652 restaurants (90.59%), while the United States follows with 434 restaurants (4.54%).
 - The remaining countries contribute relatively smaller numbers of records, with the dataset covering 15 countries in total.
 - Due to the unequal representation across countries, country-level comparisons should be interpreted with the difference in sample size in mind.
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5. City Distribution

- Within India, New Delhi, Gurgaon and Noida have the highest restaurant concentration, with **5,473, 1,118 and 1,080 restaurants**, respectively.
 - These three cities together account for **7,671 restaurants (80.32%)** of the complete dataset.
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6. Rating Analysis

- The **overall average restaurant rating is 2.67**, while **India has an average rating of 2.52**.
 - The **Average** rating category is the largest, with **3,737 restaurants**.
 - **2,148 restaurants are Not Rated**, indicating limited customer-rating coverage for a significant portion of the dataset.
 - Among the countries represented, the average ratings vary considerably; however, comparisons should be considered alongside the unequal number of records for each country.
 - The rating distribution indicates that **rating performance and customer-feedback coverage are areas for further analysis, particularly within India**.
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7. Price Range Analysis

- Price Range 1 contains 4,444 restaurants (46.53%), making it the largest price segment.
 - Price Range 2 contains 3,113 restaurants (32.59%).
 - Together, Price Ranges 1 and 2 account for 7,557 restaurants (79.12%) of the dataset.
 - Average ratings increase from 2.15 in Range 1 to 3.70 in Range 4, showing an observed relationship between price range and rating in this dataset.
 - Average cost for two varies across countries and currencies, so international cost comparisons should not be made directly without currency conversion.
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8. Online Delivery Analysis

- **2,451 restaurants (25.66%)** offer online delivery, while **7,100 (74.34%)** do not.
 - Restaurants with online delivery have an average rating of **3.25**, compared with **2.47** for those without it.
 - The data shows an observed association between online delivery availability and higher average ratings; this does not establish causation.
 - Online delivery can therefore be considered an area for further analysis and service-level improvement.
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9. Table Booking Analysis

- **1,158 restaurants (12.12%)** offer table booking, while **8,393 (87.88%)** do not.
 - Restaurants with table booking have an average rating of **3.44**, compared with **2.56** for those without it.
 - The data shows an observed association between table booking availability and higher average ratings; this does not establish causation.
 - Table booking can be further examined for restaurants where dine-in service is relevant.
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10. Currency Analysis

- **Indian Rupees (INR)** are used for **8,652 records**, matching the number of Indian restaurants.
 - Other currencies include Dollar, Pounds, Rand, AED, Brazilian Real, NZD, Turkish Lira, Botswana Pula, Indonesian Rupiah, Qatari Rial and Sri Lankan Rupee.
 - Currency usage varies across countries, so **average cost values should not be directly compared across countries without currency conversion**.
 - The Philippines–Botswana Pula mapping should be verified against the original source data before using it for financial interpretation.
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11. Key Findings

The main findings from the analysis are:

1. Strong concentration in India

India contains **90.59%** of the restaurant records, making it the main area represented in the dataset.

2. High concentration in three cities

New Delhi, Gurgaon and Noida together contain **7,671 restaurants**, or **80.32% of the complete dataset**.

3. India has an average rating of 2.52

The Indian average is below the overall dataset average of 2.67.

4. Average rating is the largest rating category

The Average category contains **3,737 restaurants**.

5. Large number of restaurants are Not Rated

There are **2,148 Not Rated records**, indicating an opportunity to examine customer-feedback coverage.

6. Lower price ranges dominate

Price Ranges 1 and 2 account for **79.12%** of the dataset.

7. Higher ratings are observed among restaurants with certain services

Restaurants with online delivery and table booking have higher average ratings in the dataset.

8. Currency standardization is required

Different currencies make direct international cost comparisons inappropriate without conversion.

12. Business Recommendations

Based on the analysis, the following areas can be considered for further business investigation:

Focus further analysis on India

Since India represents 90.59% of the dataset, further analysis can initially focus on the Indian restaurant market.

Analyze the major cities

New Delhi, Gurgaon and Noida can be examined in greater detail because they represent the largest concentration of restaurants.

Investigate the Average-rating segment

The Average rating category contains 3,737 restaurants. Further analysis can examine factors associated with these ratings, such as cuisine, price range, votes and service availability.

Examine customer feedback coverage

The dataset contains 2,148 Not Rated restaurants. Encouraging and monitoring customer ratings could provide more feedback for analysis.

Evaluate service availability

Online delivery and table booking show higher observed average ratings where available. These services can be investigated further using more detailed analysis rather than assuming a direct causal relationship.

Standardize currencies

Currency conversion should be performed before making financial comparisons between countries.

13. Dashboard

The interactive Excel dashboard summarizes the key findings from the analysis and allows the data to be explored through different dimensions.

It includes:

- KPI cards for key metrics
- Restaurant distribution by country and city
- Rating and price-range analysis
- Online delivery and table booking comparisons
- Interactive filters for **Country, City, Price Range, Rating Text, Online Delivery, and Table Booking**

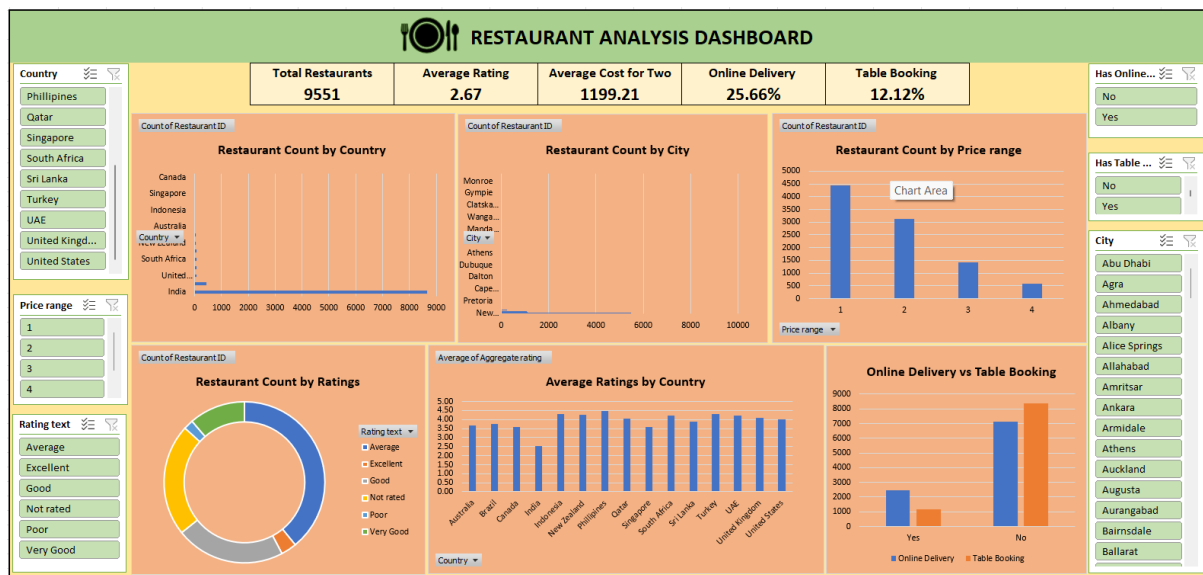


Figure 1: Restaurant Analysis Dashboard

14. Limitations

The analysis has several limitations that should be considered when interpreting the findings.

Unequal country representation

India accounts for the majority of records, while several other countries have relatively few observations. Therefore, country-level comparisons should be interpreted carefully.

Multiple currencies

The dataset contains multiple currencies, so average costs cannot be directly compared internationally without currency conversion.

Observational data

The relationships between services and ratings are observational. Higher ratings among restaurants offering online delivery or table booking do not prove that these services caused the higher ratings.

Limited time information

The available dataset does not provide sufficient time-based information for detailed analysis of rating or business performance changes over time.

15. Future Analysis

The following analyses can be performed as the next stage of the project:

- Analyze restaurant performance at the city and cuisine level.
 - Compare price range with ratings.
 - Examine service availability and customer ratings.
 - Standardize currencies for meaningful cost comparisons.
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16. Conclusion

This analysis provides an overview of restaurant distribution, ratings, pricing and service availability across the dataset.

The data is strongly concentrated in India, particularly in **New Delhi, Gurgaon and Noida**. India has an average rating of **2.52**, compared with an overall dataset average of **2.67**.

The **Average** rating category is the largest group, while **2,148 restaurants are Not Rated**, highlighting areas for further analysis of customer feedback.

Price Ranges 1 and 2 account for **79.12%** of all records, showing that the dataset is largely concentrated in the lower two price categories.

Restaurants offering online delivery and table booking have higher observed average ratings in this dataset. However, these relationships should be treated as associations rather than evidence of causation.

Overall, the analysis provides a foundation for deeper India-focused and city-level analysis, particularly around restaurant ratings, customer feedback, pricing, cuisine and service availability.